INVENTORY MANAGEMENT

Learning objectives

By the end of this topic, students should be able to:

- Describe the functions of inventories for production and operations function
- Discuss the types of stock taking of inventories
- Describe types of inventory control systems
- Explain the costs associated with managing inventories for production and operations function
- Discuss the inventory control models used to manage inventory (Economic order quantity and Just in time)

An inventory is a stock of materials used to facilitate production or to satisfy customer demands. They include raw materials, work in process and finished goods. Inventories constitute a significant proportion of the cost of production in most organizations. Manufacturing companies for example can easily be carrying inventory equivalent to between 50% and 100% of the revenue of the business. Inventory management is among the most important operations management responsibilities because inventory requires are of capital and affects the delivery of goods to customers.

Functions of Inventory

- To meet anticipated changes in demand and supply.
- To smooth production requirements. Firms that experience seasonal patterns in demand often build up inventories during pre-season periods to meet overly high requirements during seasonal periods
- To decouple operations. Historically manufacturing firms have used inventories as buffers between successive operations to maintain continuity of production that would be disrupted by events such as breakdowns of equipments. Buffers permit other operations to continue temporarily while the problem is solved. Buffers of raw materials are also used to insulate production from disruptions in deliveries from suppliers.
- To protect against stock outs. Delayed deliveries and unexpected increases in demand, increase the risk of shortage.
- To hedge against price increases. Especially when the firm suspects that a substantial price increase is about to occur. The ability to store extra goods also allows a firm to take advantage of price discounts for larger orders.

PURCHASING AND STORE KEEPING PROCEDURES

Purchasing Procedure

Purchasing is a major function of the organization's system. When purchasing materials, a number of considerations must be noted.

- The need to ensure that supplies are maintained so as to avoid the incidence of idle time or loss of future business.
- The need to minimize investment in materials so as to achieve maximum returns.

- Ensuring that purchases must be obtained at the most economic price. However, the most economic price may not necessarily be the lowest price and the lowest price should not compromise quality.
- The quality of materials produced should be maximally maintained, implying that the purchases of materials should be in line with the philosophy of Total Quality Management.
- Adequate records must be kept in order to ensure that the purchasing routine is implemented in an efficient manner.

The purchasing procedure goes through a number of steps and for accounting and control purposes, relevant documents must be prepared and kept in each stage. These stages include:

1. **Purchase Requisition.** This is raised in the response to a need that may have been expressed by any department in an organization. The purchase requisition may also be in response to the effect or regular production orders for various products or services by customers.

The purchase requisition must specify the following:-

- Quantity of materials required
- Quality of materials required.
- Required delivery date
- The responsible officer who has raised the requisition.

2. Selection of Supplier. When the purchasing department receives a duly authorized purchase requisition, a source of supply has to be selected. The purchase department maintains a list of suppliers for each type of material and selects a particular supplier after inviting tenders. The rule here is to buy the best quality at the lowest possible price after considering delivery dates and other terms of purchase.

3. Purchase Order. This is a document, which authorizes the supplier to go ahead and supply the specified quantities and qualities, at the specified date and terms. When a successful bidder has been selected, the purchasing officer prepares a purchase order. 5 copies are usually made, the original is sent to the supplier, one copy to the purchase department, then the receiving dept, accounting department and the department requisitioning for the material.

NOTE: The supplier may prepare an advise note and forward it to the company as an acknowledgement indicating the date when goods are likely to be delivered and this is normally passed on to the receiving department which has to match with the purchase order.

This note is checked to ensure that the quantity and quality of materials corresponds to the details in the purchase order.

4. Receiving and inspection of materials. When receiving the goods, the delivery note, which is sent along with the goods, should be checked properly to ensure that the quantity specification tally with the physical goods received. The goods are then sent for inspection. If on inspection, the goods are acceptable, a Goods Received Note is raised. An official responsible signs the Goods Received Note and Delivery Note and then the goods are sent to the store.

5. Payment of the supplier. At this point, the company receives an invoice from the suppliers. An invoice received is a demand order requesting the company to make payments

of goods earlier supplied. The invoices should be approved by the accounting officer and sent to the person in charge of payments.
The invoice should be checked to ensure that the following are specified.

- Purchase order reference number
- Suppliers name
- Amount of payable (VAT component)
- The date of the invoice
- Description of goods delivered and to be paid for.
- Price per unit and the total value including the details of the discounts to be given.

All supporting document should be attached and prices cross checked to make sure that all calculations are fine. A voucher is then prepared and sent for pre-auditing. A cheque in favor of the supplier is then prepared.
After payment, all documents must be stamped 'PAID'.

RECEIPT OF INVENTORY ITEMS

Duties of a Goods Receiving Section

- Recording all goods received in a ledger (goods received book- a simple list of goods received) as they arrive.
- Unpacking and checking all goods received against the purchase order especially for quantity and quality specifications.
- Returning all defective goods to the suppliers responsible. The rejection note, giving reasons as to why these goods are rejected, covers such goods. This note is raised after inspection.
- Informing the purchasing department of the receipt of all goods and this normally done by preparation of a Goods Received Note.

NOTE. After the goods receiving department has concluded its entire necessary recording, goods are passed on to the store and the storekeeper takes the charge of controlling and monitoring all material movements.

STORE KEEPING

It is also an important aspect of materials management. The storekeeper is assumed to be responsible for handling both incoming and outgoing materials. However in most cases the goods receiving department handle materials received.

Duties of a Store Keeper

These normally vary from one company to another. In some companies, the storekeeper is virtually a materials controller, keeping comprehensive records and raising all the necessary purchase requisitions. While in other organizations, a storekeeper can even be given a responsibility of receiving these items.

Whatever the organizational situation, care must be taken to give the position of a storekeeper its due importance and he is therefore expected to perform some of the following duties.

- Receiving and storing goods in good order and condition.
- Issuing materials against authorized requisitions only.

The storekeeper should never issue materials without a requisition sent by a dully-authorized person. (Except for items, which have been declared, free issue).

- Organizing materials in the store against certain relevant sets of documents. This is normally a frequent exercise for the storekeeper to conveniently assemble together all the materials in the required order using correct documents. Materials can then either be issued against a summarized materials requisition or against a well-prepared document.
- Carrying out physical stocktaking as necessary so as to verify whether the materials physically in the stores comply with the records.
- Protection of materials against theft, fire and damage and any other kind of deterioration.

STOCK TAKING

When goods are continually being moved in and out of the stores, some inaccuracies creep in and therefore to minimize the effect of these inaccuracies stocktaking is carried out.

Stocktaking is ***a physical counting*** of stock at certain periods of time to determine if the counting results correspond ***with records on stock record cards***.

Errors found are investigated and if necessary changes are made to the records to bring them in line with actual stock.

Ways of Stock Taking

1. **Annual stock taking**
   This involves physical counting of stock at the end of each accounting period. The purpose is to establish the closing stock valuation. This is normally a major exercise and it is likely to occupy a substantial number of man-hours and therefore it can disrupt production.
2. **Perpetual or continuous stock taking**
   This is a continuous method of recording store balances after each receipt and issue to facilitate regular checking for proper stock taking. The total balances represent the stock at hand. And this makes it easy to determine the inventory value at any time without physical counting.

Advantages of Perpetual Stock Taking

- The system helps in avoiding the long and costly work of physical checking of all items stocks at the end of the year.
- The Profit & Loss account, and the balance sheet can be easily prepared at interim periods as stock figures are readily available at all times.
- The system helps in keeping up to-date records.
- Discrepancies can easily be discovered and rectified.

INVENTORY CONTROL SYSTEMS

1. Re-order Level system/fixed order point system/Constant demand

It is sometimes called the **continuous review system**. ROL is the level at which a replenishment order is placed so that delivery is made when the minimum stock level is reached. The stock position is monitored after each transaction or continuously. When the stock position drops to a predetermined order point, a fixed quantity is placed on order. Since the order quantity is fixed, the time between orders will vary depending on the random nature of demand.

Advantages

- Items are ordered in economic quantities.
- Appropriate for widely differing types of inventory within the same firm.

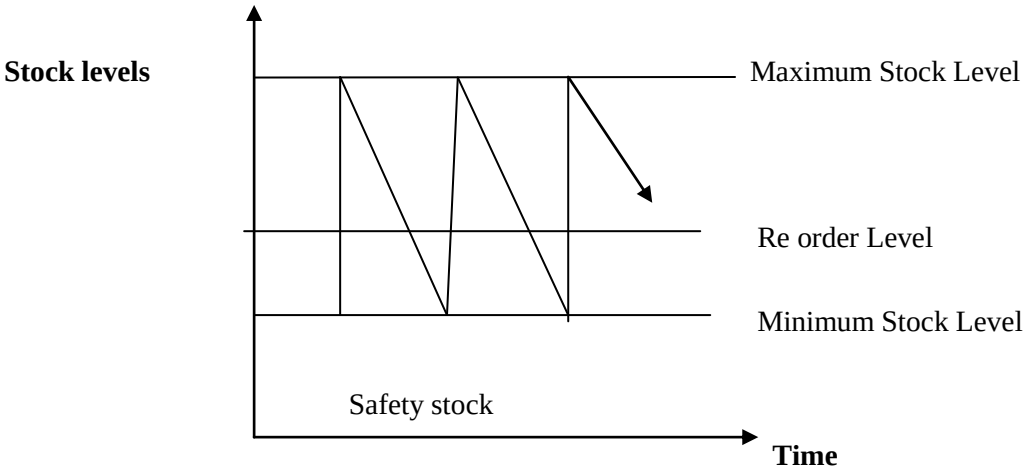
Disadvantage

- Many items may reach re-order level at the same, thus overloading the re-ordering system.

Re-Order Level system has three control levels: -

i. ***ROL.*** is the level at which a replenishment order is placed so that that delivery is made when the minimum stock level is reached
ii. **Maximum Stock Level:** This is the level above which stock should not normally rise.
iii. **Minimum Stock Level**: This is the level below which stock should not be allowed to fall. It is also known as the safety stock or buffer stock. In case stock falls below this level, stoppages in production are likely to occur.

***Diagra*m showing stock level**



2. Period Review System/ Fixed interval/variable demand

This system is sometimes called the constant cycle system. The stocks will be reviewed at regular fixed intervals to determine whether more should be ordered. An order will always be raised for the required quantity. The system has the following characteristics:

- Stock levels of all parts are reviewed at fixed intervals e.g. every 2 weeks.
- Where necessary a replenishment order is issued.
- The quantity of the replenishment order is not a previously calculated EOQ, but is based upon the likely demand until the next review, the present stock level and lead-time.
- The replenishment order quantity seeks to bring stocks up to a predetermined level.
- The effect of the system is to order variable quantities at fixed intervals as compared with the re-order level system, where fixed quantities are ordered at variable intervals.

Advantages

- All stocks are reviewed periodically so that there is more chance of obsolete items being eliminated.
- The problem of many items reaching re order level at the same time is avoided.

3. ABC inventory management system

This system was developed by Vilfredo Pareto in 1906 who observed that few items in any group may constitute a significant proportion of the entire group. At that time he was concerned that a few individuals in the economy seemed to earn most of the income. It can also be observed that a few products in the firm result in most of the sales and in some organizations; a few people do most of the work.

Based on the above principle, in inventories, a few items can account for most of the inventory value thus; one can manage few inventory items intensively and control most of the inventory value. In inventory work, Items can be divided into three classes: A, B and C often called the **ABC analysis or the 80-20 rule**. With this classification,

- Class A contains 20% of total inventory items with 80% of total inventory value.
- Class B contains 30% of total inventory items with 15% of the total inventory value.
- Class C contains 50% of the inventory items but 5% of the total inventory value.

A tight control system is needed for closely monitoring class A items. There is a need for continuous reviews of stock, less safety stock and close attention to record accuracy.

COSTS ASSOCIATED WITH MANAGING INVENTORY

Keeping and managing inventory in an organization involve costs. These costs can be classified as:

1. Ordering costs/ acquisition costs. Cost associated with ordering a batch or lot of items. It does not depend on the number of items ordered but assigned to the entire batch. It includes:

i. Administrative costs associated with paper work
ii. Costs of typing a purchase order
iii. Costs associated with contacting the supplier and dispatching the order.
iv. Cost of shipping/transportation's (delivery costs)
v. Cost of receiving and inspection
vi. Costs associated with placing the materials in the stores.

2. **Holding costs/ carrying costs**

This is the cost of holding stock. Costs associated with keeping items in the inventory for a period of time. The larger the quantity, the higher the costs.

Holding/ Carrying costs include: -

i. Cost of capital locked up in the inventory i.e. opportunity costs.
ii. Storage costs e.g. renting, cooling, lighting, heating etc.
iii. Insurance costs
iv. Taxes
v. Obsolescence
vi. Damage and deterioration

3. **Stock out costs**

These are the costs associated with running out of stock. These include: -

- Lost contribution through the lost sale caused by the stock out.
- Loss of future sales because customers go elsewhere.
- Cost of production stoppage (idle resources)
- Labor frustration over stoppages.
- Contractual liability
- Damage of company reputation
- Costs of failure to fulfill emergency orders

4. **Item purchase cost.**

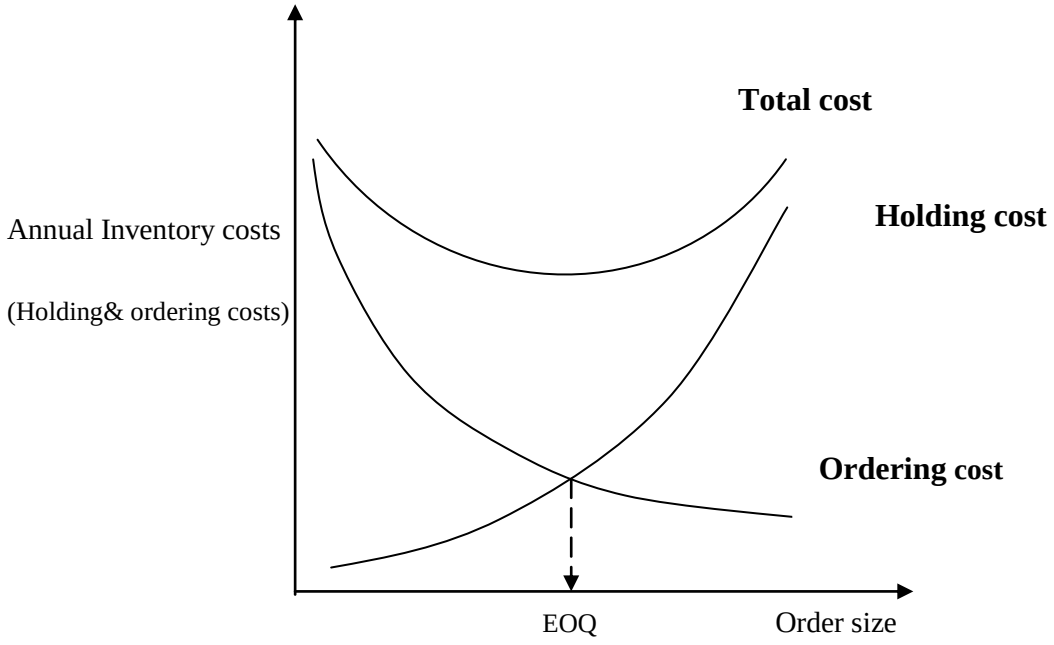
It is the cost of buying or producing individual inventory items. It is usually expressed as a cost per unit multiplied by quantity procured or produced. Sometimes item cost is discounted if enough units are purchased at one time.

INVENTORY CONTROL MODELS

1. Economic Order Quantity Analysis

It was developed in 1915 by F.W. Harris and gained wide use in industry through the efforts of a consultant named Wilson. Thus the EOQ formula is often called the Wilson EOQ even though it was developed by Harris. The EOQ model is widely used to decide optimum ***order size of stock***, which will minimize the costs of ordering and holding stock. It is a deterministic type of model, which assumes that all parameters (factors) are known with certainty..

EOQ MODEL



The derivation of the EOQ model is based on the following assumptions

Assumptions

- Ordering and holding costs are known with certainty.
- The lead time is constant and known. The lead time from order placement to order delivery, is therefore always a fixed number of days.
- No stock outs are allowed. Since demand and lead time are constant, one can determine exactly when to order materials to avoid stock outs.
- The rate of demand is constant and known. The demand is assumed to continue into the indefinite future.
- A specific cost structure is used as follows: The unit item cost is constant, and no discounts are given for large purchases. The carrying/holding cost depends linearly on the average inventory level. There is a fixed ordering cost for each lot, which is independent of the number of items in the lot.

- The item is a single product; there is no interaction with other products.

- Economic order size (quantity) is given by the formula.

$$Q = \sqrt{2\frac{D\,C_o}{C_h}}$$

Where:
D = annual demand/ usage rate in units
C_o = cost of placing an order
C_h = holding cost per unit.
Q = Economic order quantity in units

Other equations:

- Average stock = $\frac{Q}{2}$

  Where ***Q*** represents order size in units ***OR Economic*** Order Quantity

- Number of orders per annum = $\frac{D}{Q}$

  That is; total annual demand divided by the order size in units

- Annual ordering cost is the product of the ***number of orders per year*** and the ***cost per order*** ***That is; ORDERING COST =*** $C_O * \frac{D}{Q}$

- Annual holding cost is the product of the ***annual cost of holding one unit * the average stock that is HOLDING COST =.*** $C_h * \frac{Q}{2}$

- Total cost is the sum of total holding and ordering costs = $C_h * \frac{Q}{2} + C_O * \frac{D}{Q}$

Example one
Assume a product with;
Annual demand/usage rate of 2400 units
Ordering cost is 100/= per order
Annual carrying cost is 12/=. Per unit of stock
Required: Calculate the EOQ

$$EOQ = 2\frac{\text{x 2400 x 100}}{12} = \underline{\text{200 units}}.$$

Example two
Assume 2 products A & B.
Min usage = 50 units per week each.
Max usage = 150 units per week each
Average usage = 100 units per week each.
ROQ = A = 600 units, B = 1000 units
Lead-time = A = 4-6 weeks, B = 2-4 weeks.
Annual demand/usage rate for a product is 2400 units
Ordering cost is 100/= per order
Annual carrying cost is 12/=. Per unit of stock

Required: Calculate the various stock levels

Solution:

1. **ROL**
   ROL = Max usage rate x max lead-time
   A = 150 x 6 = 900 units
   B = 150 x 4 = 600 units.
2. **Min stock level**
   = ROL – (average usage rate x average lead-time)
   A = 900 – (100 x 5) = 400 units
   B = 600 – (100 x 3) = 300 units

3. **Max stock level**
   = ROL + ROQ – (Min usage rate x min lead-time)
   A = 900 + 600 – (50 x 4) = 1300 units
   B = 600 + 1000 - (50 x 2) = 1500 units

4. **Average stock level**
   = ½ (min level + max level) or $\frac{Q}{2}$

   A = ½ (400 + 1300) = 850 units
   B = ½ (300 + 1500) = 900 units

5. **Calculate the EOQ**

$$EOQ = \frac{2 \times 2400 \times 100}{12} = \underline{200 \text{ units}}.$$

Illustration III
A company requires 1000 units of material X per month. The cost per order is $ 30 regardless of the size of the order. The holding costs are $ 2.88 per unit per annum. Calculate the EOQ.

Illustration IV
Fast food restraint has a local retail outlet that uses 730 crates of soda annually. Ordering costs are $12 and carrying costs are 30% of average purchase cost of soda and a crate of soda costs $8.Determine the economic order quantity?

Revision Questions

1. Wax ltd requires 8000 units of material Z semi-annually for its operational activities and the cost per order is $20 regardless of the order size. The holding costs per unit are $4 per annum.
Required: Calculate the Economic order quantity, total annual holding costs and total annual ordering costs for the company that has engaged in the above transactions.

2. The following information relates to product X
   - Minimum usage = 100 units per month
   - Maximum usage = 200 units per month
   - Re-order quantity = 500 units

- Lead time = 2-4 months

Required: Determine:

i. Re-order level
ii. Minimum stock level
iii. Maximum stock level
iv. Average stock level

2. Just In Time (JIT) Technique

Items of very high value, that are frequently used, it makes sense to attempt to keep low or possibly no inventory of these items and instead get very frequent deliveries possibly on daily basis. The objective is to avoid tying lots of money in stock (working capital) the organisation will need to find extremely reliable suppliers who are JIT oriented. JIT is based on "***PULL***" concept, whereby suppliers wait for a signal from their customers before they deliver or replenish.

Advantages/Benefits

- Reduced or zero inventory holding costs
- Lower working capital costs
- Fresher products
- Improved quality due to fresher products
- More productive usage of space

Disadvantages/Limitations

- Suppliers may not be capable
- Suppliers will require greater contractual security
- Higher transport and delivery costs
- The organization is highly dependent upon its suppliers
- If demand suddenly increases, the organization may not be capable of responding to the change in demand.
- It requires more deliveries thus potentially more traffic, pollution and congestion

Revision questions

1. Explain the reasons why it is important to have inventories in the production and operations function of a firm
2. Explain the difference between annual and perpetual stock taking in inventory management
3. Describe the difference between re-order level and periodical review systems of controlling inventory levels in production and operations management
4. Describe the use of ABC analysis in managing levels of inventories required for production and operations function in an organization.
5. Explain various costs associated with managing inventory levels in an organization
6. Describe the assumptions in the application of economic order quantity model of inventory management
7. Suppose a store's manager controls a ware house that distributes a certain type of breakfast food to retailers. The breakfast food has the following characteristics: Average demand = 200 cases per day, ordering costs = $20 per order, Holding costs per case per annum = 20% of purchase cost and the purchase costs = $10 per case. Assuming that a continuous review system is used and the warehouse is open 250

days in a year, determine the economic order quantity and the total of ordering and holding costs.

8. Describe the use of Just in Time technique of inventory management and explain its benefits and limitations in production and operations management.